

MUNISING, MI, USA

WMO: 720198

Lat: 46.417N

Lon: 86.650W

Elev: 187

StdP: 99.10

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 54813

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.7	-17.2	-24.5	0.4	-18.7	-22.3	0.5	-16.8	12.8	-7.9	12.0	-6.9	4.1	180	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	29.7	21.1	27.7	20.1	26.0	19.4	22.5	27.1	21.5	26.0	20.5	24.5	4.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	16.1	25.1	19.9	14.9	24.0	18.9	14.0	23.0	67.3	26.8	63.4	25.9	59.8	24.6	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	9.9	8.7	DB	-23.5	32.9	3.4	1.5	-25.9	34.0	-27.9	34.9	-29.9	35.7	-32.3	36.7	(4)
(5)			WB	-23.7	24.7	3.3	1.3	-26.0	25.7	-27.9	26.4	-29.8	27.1	-32.1	28.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-6.5	-6.7	-1.9	3.8	9.9	14.9	18.9	18.7	15.5	8.6	2.4	-3.4	(6)	
	DBStd	10.32	5.50	5.43	5.83	4.72	5.22	4.57	3.87	3.33	4.35	4.27	4.95	4.99	(7)	
	HDD10.0	2355	511	468	373	196	68	7	0	0	4	81	232	415	(8)	
	HDD18.3	4565	769	701	628	436	268	122	42	36	109	303	477	673	(9)	
	CDD10.0	987	0	0	3	10	65	156	276	269	168	36	4	0	(10)	
	CDD18.3	156	0	0	0	0	6	20	59	47	22	1	0	0	(11)	
	CDH23.3	1102	0	0	2	2	77	177	414	304	122	5	0	0	(12)	
	CDH26.7	277	0	0	1	0	21	43	116	68	27	0	0	0	(13)	
(14) Wind	WSAvg	4.0	4.5	4.5	4.4	4.3	3.8	3.2	3.4	3.4	3.8	4.2	4.4	4.2	(14)	
(15) Precipitation	PrecAvg	909	76	50	53	55	74	79	82	83	100	105	79	84	(15)	
	PrecMax	1216	206	139	129	106	157	189	175	185	179	234	174	205	(16)	
	PrecMin	654	36	11	13	7	29	12	14	5	15	27	19	17	(17)	
	PrecStd	154	37	27	28	23	35	43	46	37	39	49	31	36	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	7.3	18.3	21.7	29.2	30.6	32.2	31.3	30.1	23.8	17.3	8.5	(19)	
		MCWB	3.5	4.5	13.9	12.6	19.2	21.3	22.5	22.1	22.2	16.9	12.2	6.4	(20)	
	2%	DB	3.8	4.4	11.4	18.1	25.4	27.7	29.8	28.5	26.6	20.2	13.4	5.6	(21)	
		MCWB	2.1	1.9	7.1	10.5	17.2	19.5	21.3	20.7	19.4	15.5	9.8	4.1	(22)	
	5%	DB	2.3	2.1	8.1	14.8	21.7	25.2	27.6	26.6	24.1	17.6	11.1	4.0	(23)	
		MCWB	1.1	0.5	4.6	8.4	15.2	18.5	20.3	20.2	18.8	14.2	8.3	2.6	(24)	
	10%	DB	1.1	0.7	5.5	11.5	18.6	22.7	25.6	24.8	22.3	15.3	9.2	2.9	(25)	
		MCWB	0.0	-0.8	2.8	6.9	12.9	17.2	19.7	19.5	18.0	12.6	6.8	1.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	5.2	14.1	14.1	21.1	22.9	24.6	23.6	23.1	18.7	13.1	7.1	(27)	
		MCDB	4.9	7.1	18.5	19.5	27.2	28.9	30.0	28.3	28.3	20.1	16.6	8.2	(28)	
	2%	WB	2.4	2.4	7.9	11.7	18.7	20.6	22.5	22.3	21.1	16.4	10.6	4.6	(29)	
		MCDB	3.5	3.9	10.8	16.2	23.6	26.0	26.9	26.5	24.8	20.4	12.9	5.4	(30)	
	5%	WB	1.2	0.6	5.0	9.0	16.3	19.3	21.5	21.2	19.8	14.6	8.6	2.9	(31)	
		MCDB	2.2	1.8	7.8	14.1	20.7	23.7	25.9	25.2	23.0	17.5	10.7	3.8	(32)	
	10%	WB	0.0	-0.8	2.9	6.8	13.9	17.9	20.4	20.2	18.6	12.6	6.9	1.7	(33)	
		MCDB	1.1	0.7	5.2	11.5	17.8	21.9	24.5	23.8	21.6	15.0	9.0	2.8	(34)	
(35) Mean Daily Temperature Range	MDBR		6.4	7.7	8.1	8.4	9.9	10.0	9.8	9.6	9.5	7.7	6.6	6.1	(35)	
	5% DB	MCDBR	6.2	8.6	11.1	15.2	15.1	13.9	12.5	12.2	11.5	11.1	9.4	6.8	(36)	
		MCWBR	5.5	6.7	7.4	9.2	8.6	7.7	6.5	6.3	6.1	7.3	6.4	5.9	(37)	
	5% WB	MCDBR	5.9	8.0	9.8	13.6	13.9	11.8	10.8	9.9	9.7	10.1	8.5	6.1	(38)	
		MCWBR	5.5	6.5	7.0	8.9	8.8	7.5	6.3	5.9	5.9	7.3	6.3	5.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.268	0.281	0.312	0.347	0.375	0.385	0.390	0.376	0.361	0.333	0.303	0.269	(40)	
	taud		2.427	2.376	2.375	2.377	2.368	2.372	2.376	2.415	2.458	2.516	2.518	2.473	(41)	
	Ebn at Noon		845	900	917	910	891	881	870	871	854	830	796	804	(42)	
	Edh at Noon		68	87	102	112	118	118	116	108	94	77	63	59	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.07	1.92	3.35	4.45	5.45	5.90	6.03	5.10	3.78	2.24	1.19	0.84	(44)	
	RadStd		0.08	0.26	0.31	0.46	0.39	0.45	0.31	0.33	0.26	0.26	0.15	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page